

Gil Tal, Ph.D.

Curriculum Vitae

Director, Electric Vehicle Research Center
Director, STEPS+ Research Consortium
Adjunct Professor, Department of Environmental Science and Policy
Institute of Transportation Studies, University of California, Davis
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EDUCATION

Ph.D., Transportation Technology and Policy, University of California, Davis	2008
Postdoctoral Researcher, University of California, Berkeley	2008 to 2010
M.A., Geography and Environmental Policy, Hebrew University of Jerusalem	

ACADEMIC APPOINTMENTS

Adjunct Professor, Department of Environmental Science and Policy, UC Davis	Effective July 2026
Associate Adjunct Professor, Department of Environmental Science and Policy, UC Davis	2023 to 2026
Director, Electric Vehicle Research Center, UC Davis Institute of Transportation Studies	2018 to present
Director, STEPS+ Research Consortium, UC Davis Institute of Transportation Studies	2022 to present
Associate Professional Researcher, UC Davis Institute of Transportation Studies	2018 to 2023
Assistant Professional Researcher, UC Davis Institute of Transportation Studies	2014 to 2018

HONORS AND AWARDS

Graduate Program Advising and Mentoring Award, UC Davis	2021
Standing Committee on Alternative Fuels and Technologies, Transportation Research Board, National Academies	2020 to 2028

RESEARCH GRANTS (PRINCIPAL INVESTIGATOR UNLESS NOTED)

Active

EV Policy, Critical Minerals, and Environmental Justice Programs. Heising-Simons Foundation. \$250,000 (PI)	2024 to 2026
Remote DCFC Reliability and Downtime Detection Tool. Caltrans. \$200,000 (PI)	2024 to 2026
State of the Zero-Emission Vehicle Secondary Market and Accessibility Impacts in California's Underserved Communities. California Air Resources Board. \$500,000 (PI)	2023 to 2026
Measuring Charging Infrastructure Reliability in California. California Energy Commission. \$1,173,568 (PI)	2023 to 2026
STEPS+ EV Center. STEPS Sponsors Consortium. \$1,250,000 (PI)	2023 to 2026

Recently Completed

Implementing LDV and HDV ZEV Regulations Around the World. ClimateWorks Foundation. \$401,000 (PI)	2023 to 2025
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EV Center Policy Engagement, Environmental Justice Leaders, and Critical Minerals Supply and Demand. Heising-Simons Foundation. \$230,000 (PI) 2023 to 2024

Electrification of Transport: Challenges and Opportunities for the U.S. Mexico Transportation Industry. UC Office of the President. \$127,661 (PI) 2023 to 2024

Measuring the Emissions and Socioeconomic Benefits of CARB's Incentive and Regulatory Programs. UC Berkeley subaward. \$181,127 (PI) 2021 to 2023

Next10 EV Infrastructure Business Models. Next10. \$90,000 (PI) 2022 to 2023

Virginia Expansion of the GIS Toolbox. National Rural Electric Cooperative Association. \$55,000 (PI) 2021 to 2023

Co-Investigator

The Center for Emissions Reduction, Resiliency, and Climate Equity in Transportation (CERRCET). U.S. Department of Transportation. \$1,750,000 total (Co-PI; PI: Kari Watkins) 2024 to 2025

Total funding as Principal Investigator during the 2023 to 2025 review period: \$3.2 million across six new awards.

PEER-REVIEWED JOURNAL PUBLICATIONS

2026

- Tal, G., & Kurani, K. (2026). Zero-emission vehicles are entering the used market: What does this mean for California? *Institute of Transportation Studies, UC Davis Working Paper Series*.
- Chakraborty, D., Konstantinou, T., Gutierrez Lopez, J. B., & Tal, G. (2026). Understanding today's used car buyers can help grow the market for used electric vehicles. *Institute of Transportation Studies, UC Davis Working Paper Series*.

2025

- Plötz, P., & Tal, G. (2025). Regulate reality in vehicle emission policy. *Communications Earth & Environment*, 7(1), 76.
- Busch, P., Parés, F., Tal, G., & Kendall, A. (2025). Who will have enough? Battery mineral demand and sufficiency in vehicle producing countries. *Environmental Science & Technology*, 59(45), 24392 to 24403.
- Ramadoss, T. V., Jenn, A., Davis, A. W., & Tal, G. (2025). Why buy multiple electric vehicles? The roles of vehicle ownership, charging access, and energy choices in California. *Energy Research & Social Science*, 129, 104391.
- Tayarani, H., Rabinowitz, A., Jenn, A., & Tal, G. (2025). Assessment of vehicle-grid integration profitability subject to real-world driver behavior and electricity tariff. *Energy*, 139302.
- Chandra, M., Busch, P., Parés Olguín, F., & Tal, G. (2025). Paths of progress: Forecasting global electric vehicle demand amidst demographic and economic growth. *Transportation Research Part D: Transport and Environment*, 147, 104928.
- Ramadoss, T. V., Lee, J. H., Davis, A. W., Hardman, S., & Tal, G. (2025). Classifying electric vehicle adopters and forecasting progress to full adoption. *npj Sustainable Mobility and Transport*, 2(1), 32.
- Weekx, S., Tal, G., & Vanhaverbeke, L. (2025). EV charging habits and their impact on public charging infrastructure usage. *Transportation Research Part D: Transport and Environment*, 143, 104730.
- Kwon, K., & Tal, G. (2025). Evaluating spatial equality of DC fast charging infrastructure: A case study of California. *Journal of Urban Planning and Development*, 151(3), 05025026.
- Kwon, K., Li, X., & Tal, G. (2025). Exploring new tools for modeling home and commute charging demand: A California case study. *Transportation Planning and Technology*, 1 to 16.

Hardman, S., Aksen, J., Chakraborty, A., Figenbaum, E., Helveston, J. P., Hoogland, K., Jenn, A., Jensen, A. F., Jochem, P., Plötz, P., Pontes, J., Refa, N., Sprei, F., Sugihara, C., Tal, G., & Williams, B. (2025). Demand-side challenges and research needs on the road to 100% zero-emissions vehicles. *Progress in Energy*, 7(2), 022001.

Jenn, A., Chakraborty, A., Hardman, S., Hoogland, K., Sugihara, C., Tal, G., Helveston, J. P., Rich, J., Jochem, P., Plötz, P., Sprei, F., Williams, B., Aksen, J., Figenbaum, E., Pontes, J., & Refa, N. (2025). Supply-side challenges and research needs on the road to 100% zero-emissions vehicle sales. *Progress in Energy*, 7(2), 022002.

2024

Hardman, S., Shafaeen, M., & Tal, G. (2024). Identifying mass market adoption in the transition to electric vehicles. *Environmental Research Letters*, 19(11), 110055.

Parés Olguín, F., Busch, P., Chandra, M., & Tal, G. (2024). Shifting manufacturing: Electric vehicle supply strategy using the model for international EV trade. *Journal of Cleaner Production*, 484, 144357.

Busch, P., Parés, F., Chandra, M., Kendall, A., & Tal, G. (2024). Future of global electric vehicle supply chain: Exploring the impact of global trade on electric vehicle production and battery requirements. *Transportation Research Record*. (Introducing the MONET model.)

Gamage, T. D., Jenn, A. T., & Tal, G. (2024). Novel methodology to measure the reliability of public DC fast charging stations. *Transportation Research Record*, 2678(11), 1440 to 1453.

Chakraborty, D., Davis, A. W., & Tal, G. (2024). The cost of aggressive electrification targets: Who bears the burden without mitigating policies? *Transportation Research Interdisciplinary Perspectives*, 23, 101006.

Weekx, S., Tal, G., & Vanhaverbeke, L. (2024). How many more public charging stations do we need? A data-driven approach considering charging station overflow dynamics. *Transportation Research Record*.

2023

Lee, J. H., Cho, M., Tal, G., & Hardman, S. (2023). Do plug-in hybrid adopters switch to battery electric vehicles (and vice versa)? *Transportation Research Part D: Transport and Environment*, 119, 103752.

Gamage, T., Tal, G., & Jenn, A. T. (2023). The costs and challenges of installing corridor DC fast chargers in California. *Case Studies on Transport Policy*, 11, 100969.

Chakraborty, D., Lee, J. H., Chakraborty, A., & Tal, G. (2023). To adopt rooftop solar or not along with electric vehicles? Exploring the factors influencing co-adoption decisions among electric vehicle owners in California. *The Electricity Journal*, 36(7), 107315.

2022

Chakraborty, D., Bunch, D. S., Brownstone, D., Tal, G., & Xu, B. (2022). Plug-in electric vehicle diffusion in California: Role of exposure to new technology at home and work. *Transportation Research Part A*, 156, 133 to 151.

Mandev, A., Plötz, P., Sprei, F., & Tal, G. (2022). Empirical charging behavior of plug-in hybrid electric vehicles. *Applied Energy*, 321, 119293.

Hardman, S., Chakraborty, D., & Tal, G. (2022). Estimating the travel demand impacts of semi-automated vehicles. *Transportation Research Part D: Transport and Environment*, 107, 103311.

Raghavan, S. S., & Tal, G. (2022). Plug-in hybrid electric vehicle observed utility factor: Why the observed electrification performance differs from expectations. *International Journal of Sustainable Transportation*, 16(2), 105 to 136.

Chakraborty, D., Hardman, S., & Tal, G. (2022). Integrating plug-in electric vehicles into household fleets: Factors influencing miles traveled by PEV owners in California. *Travel Behaviour and Society*, 26, 67 to 83.

Tsoi, K. H., Loo, B. P. Y., Tal, G., & Sperling, D. (2022). Pioneers of electric mobility: Lessons about transport decarbonisation from two Bay Areas. *Journal of Cleaner Production*, 330, 129866.

Karanam, C. V., Davis, A., Sugihara, C., Sutton, K., & Tal, G. (2022). From shifting gears to changing modes: The impact of driver inputs on plug-in hybrid electric vehicle energy use and emissions. *Transportation Research Interdisciplinary Perspectives*, 14, 100597.

2021

Hardman, S., & Tal, G. (2021). Understanding discontinuance among California's electric vehicle owners. *Nature Energy*, 6(5), 538 to 545.

Muratori, M., Alexander, M., Arent, D., Bazilian, M., Dede, E. M., Farrell, J., Gearhart, C., Greene, D., Jenn, A., Keyser, M., et al., with Tal, G. (2021). The rise of electric vehicles: 2020 status and future expectations. *Progress in Energy*, 3(2), 22002.

Benoliel, P., Jenn, A., & Tal, G. (2021). Examining energy uncertainty in battery bus deployments for transit agencies in California. *Transportation Research Part D: Transport and Environment*, 98, 102963.

Raghavan, S. S., & Tal, G. (2021). Technology implications of electromobility on household travel emissions. *Transportation Research Part D: Transport and Environment*, 94, 102792.

Sugihara, C., Sutton, K., Davis, A., Karanam, V., & Tal, G. (2021). From sport to eco: A case study of driver inputs on electric vehicle efficiency. *Transportation Research Part F: Traffic Psychology and Behaviour*, 82, 412 to 428.

Hamza, K., Chu, K. C., Favetti, M., Benoliel, P. K., Karanam, V., Laberteaux, K. P., & Tal, G. (2021). Comparisons of real-world vehicle energy efficiency with dynamometer-based ratings and simulation models. *World Electric Vehicle Journal*, 12(4), 161.

Karanam, V. C., & Tal, G. (2021). Emission implications of plug-in hybrid electric vehicles through an empirical exploration of engine starts. *Transportation Research Record*.

2020

Jenn, A., Lee, J. H., Hardman, S., & Tal, G. (2020). An in-depth examination of electric vehicle incentives: Consumer heterogeneity and changing response over time. *Transportation Research Part A: Policy and Practice*, 132, 97 to 109.

Chakraborty, D., Hardman, S., & Tal, G. (2020). Why do some consumers not charge their plug-in hybrid vehicles? Evidence from Californian plug-in hybrid owners. *Environmental Research Letters*, 15(8), 084030.

Lee, J. H., Chakraborty, D., Hardman, S., & Tal, G. (2020). Exploring electric vehicle charging patterns: Mixed usage of charging infrastructure. *Transportation Research Part D: Transport and Environment*, 79, 102249.

Ji, W., & Tal, G. (2020). Scenarios for transitioning cars from internal combustion engine vehicles to battery electric and plug-in hybrid electric vehicles using household-level GPS travel data. *Transportation Research Part D*, 88, 102555.

2019

Chakraborty, D., Bunch, D. S., Lee, J. H., & Tal, G. (2019). Demand drivers for charging infrastructure: Charging behavior of plug-in electric vehicle commuters. *Transportation Research Part D: Transport and Environment*, 76, 255 to 272.

Lee, J. H., Hardman, S., & Tal, G. (2019). Who is buying electric vehicles in California? Characterizing early adopter heterogeneity and forecasting market diffusion. *Energy Research & Social Science*, 55, 218 to 226.

Hardman, S., Berliner, R., & Tal, G. (2019). Who will be the early adopters of automated vehicles? Insights from a survey of electric vehicle owners in the United States. *Transportation Research Part D: Transport and Environment*, 71, 248 to 264.

- Berliner, R. M., Hardman, S., & Tal, G. (2019). Uncovering early adopter's perceptions and purchase intentions of automated vehicles: Insights from early adopters of electric vehicles in California. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 712 to 722.
- Canepa, K., Hardman, S., & Tal, G. (2019). An early look at plug-in electric vehicle adoption in disadvantaged communities in California. *Transport Policy*, 78, 19 to 30.
- Hardman, S., Lee, J. H., & Tal, G. (2019). How do drivers use automation? Insights from a survey of partially automated vehicle owners in the United States. *Transportation Research Part A: Policy and Practice*, 129, 246 to 256.

Selected Earlier Publications (2010 to 2018)

- Hardman, S., Jenn, A., Tal, G., Axsen, J., Beard, G., Daina, N., Figenbaum, E., Jakobsson, N., Jochem, P., Kinnear, N., Plötz, P., Pontes, J., Refa, N., & Sprei, F. (2018). A review of consumer preferences of and interactions with electric vehicle charging infrastructure. *Transportation Research Part D*, 62, 508 to 523.
- Hardman, S., & Tal, G. (2018). Who are the early adopters of fuel cell vehicles? *International Journal of Hydrogen Energy*, 43(37), 17857 to 17866.
- Hardman, S., Chandan, A., Tal, G., & Turrentine, T. (2017). The effectiveness of financial purchase incentives for battery electric vehicles: A review of the evidence. *Renewable & Sustainable Energy Reviews*, 80, 1100 to 1111.
- Wang, Y., Sperling, D., Tal, G., & Fang, H. (2017). China's electric car surge. *Energy Policy*, 102, 486 to 490.
- Tal, G., & Nicholas, M. (2016). Exploring impact of the federal tax credit on the plug-in vehicle market. *Transportation Research Record*, 2572, 95 to 102.
- Tal, G., Nicholas, M. A., Davies, J., & Woodjack, J. (2014). Charging behavior impacts on electric vehicle miles traveled: Who is not plugging in? *Transportation Research Record*, 2454(1), 53 to 60.
- Tal, G., & Cohen-Blankshtain, G. (2011). Understanding the sources of overestimation bias in studies of travel demand management policies: Optimism bias versus scientific skepticism. *Transportation Research Part A: Practice and Policy*, 45(5), 389 to 400.
- Tal, G., & Handy, S. L. (2010). Travel behavior of immigrants: Analysis of 2001 National Household Travel Survey. *Transport Policy*, 17(2), 85 to 93.

Career publication summary: Over 200 publications in Google Scholar, including more than 60 peer-reviewed journal articles. Total citations: approximately 7,400 (Google Scholar). h-index: 38. i10-index: 80.

EDITED VOLUMES AND BOOK CHAPTERS

- Contestabile, M., Tal, G., & Turrentine, T. (Eds.). (2020). *Who's driving electric cars: Understanding consumer adoption and use of plug-in electric cars*. Springer Nature, Switzerland.
- Kuby, M. J., Martinez, A. S., Kelley, S. B., & Tal, G. (2023). Hydrogen station location analysis and optimization: Advanced models and behavioral evidence. In *Hydrogen Economy* (pp. 315 to 380). Academic Press.
- Tal, G., Kurani, K., Jenn, A., Chakraborty, D., Hardman, S., & Garas, D. (2020). PEVs in the U.S.: Policy evolution and behavioural perspectives. In M. Contestabile, G. Tal, & T. Turrentine (Eds.), *Who's driving electric cars*. Springer Nature.
- Mokhtarian, P., & Tal, G. (2013). Impacts of ICT on travel behavior: A tapestry of relationships. In J. Shaw, J. P. Rodrigue, & T. Notteboom (Eds.), *Transport Handbook* (p. 241). Sage Publication, London.

LEGAL AND LEGISLATIVE SERVICE

Amicus brief, U.S. Supreme Court (No. 24-7), in support of respondents on clean transportation policy	March 2025
Amicus brief, U.S. Court of Appeals, D.C. Circuit (No. 24-1129), on EPA clean vehicle rules	January 2025
Amicus brief, U.S. Court of Appeals, D.C. Circuit (No. 24-1087), on EPA clean vehicle rules	December 2024
Briefing, California Council on Science and Technology, on preparing California's grid for new load sources at scale	June 2025
Capitol Hill Research Briefing on federal policy opportunities for improving the U.S. EV charging ecosystem	January 2025
Testimony, California Assembly Standing Committee on Utilities and Energy	February 2024
Capitol Hill Research Briefing for the National Council on Sustainable Transportation	January 2024

SELECTED INVITED TALKS AND CONFERENCES

2025 to 2026

CERAWeek 2026 panel: "The Five-Minute Challenge: Ultra-Fast Charging vs. Battery Swapping," Houston	March 2026
UCLA Lake Arrowhead Symposium, panel discussion on the future of new mobility	April 2026
Accelerating the Transition Conference, SF Climate Week, panel with Dan Sperling, Nancy Skinner, Paul Stith, Xantha Bruso	April 2026
UN Environment Programme Drive Electric Advisory Group, Nairobi, Kenya	June 2025
ICCT 4th Transatlantic Decarbonization Summit, London	June 2025
Electric Vehicle Conference 2025, Stuttgart, Germany (Conference Organizer)	May 2025
Workshop: Designing Supply Side Vehicle Regulations, UC Davis	May 2025
SAE Government and Industry Meeting, Washington DC	January 2025
Mobility Global Forum, Detroit Auto Show	January 2025
20th Asilomar Conference on Transportation, Environment, and Energy Policy (Co-Organizer)	2025

2023 to 2024

17th International Conference on Travel Behaviour Research, Vienna, Austria	July 2024
ICCT 3rd Transatlantic Transportation Decarbonization Summit, Estérel, Québec	June 2024
Industry Studies Association Conference, Sacramento, CA	June 2024
Distributed Energy Resources Membership Advisory Group, Milwaukee, WI	June 2024
STEPS+ Spring Research Symposium, UC Davis (Organizer)	May 2024
Subaru EV Workshop, Philadelphia (Keynote)	March 2024
STEPS+ Symposium, UC Davis (Organizer)	November 2023
University of New South Wales, Sydney, Australia	October 2023
UN Drive Electric Advisory Group Working Meeting, Costa Rica	September 2023

TEACHING

Annual undergraduate course in the Environmental Policy, Analysis and Planning major, alternating between two key track courses. Combined annual enrollment of more than 160 students. Mean teaching evaluation: 4.2 of 5.0.

ESP/ECI 163, Energy and Environmental Aspects of Transportation
ECI 165, Transportation Policy

Fall 2024, ongoing
Fall 2023, ongoing

GRADUATE MENTORING

Recipient, Graduate Program Advising and Mentoring Award, UC Davis (2021). Admissions Adviser, Transportation Technology and Policy Graduate Group (2017 to present).

During the most recent two-year period, advised four Ph.D. students, five Master's students, six postdoctoral researchers, and two visiting scholars, and mentored four undergraduates through research projects. Six former Ph.D. students have completed their degrees and now hold positions in academia, government, and industry in the United States and internationally. Has served on more than 35 thesis and dissertation committees across multiple programs.

Current and Recent Doctoral Advisees

- Trisha Varshini Ramadoss, Ph.D., Chair (in progress)
- Hanif Tayarani, Ph.D., Chair (in progress)
- Tisura Dasith Gamage, Ph.D., Co-Chair (in progress)
- Anya Rose Robinson, Ph.D., Chair (in progress)
- Lu Dong, Ph.D., Chair (in progress)
- Peter Benoliel, Ph.D., Chair (awarded)
- Claire Sugihara, Ph.D., Committee Member (awarded)

PROFESSIONAL AND UNIVERSITY SERVICE

Director, Electric Vehicle Research Center, UC Davis ITS (managing 30+ researchers, staff, and students) 2018 to present

Director, STEPS+ Research Consortium (5 research centers, 25 sponsoring organizations) 2022 to present

Admissions Adviser, Transportation Technology and Policy Graduate Group, UC Davis 2017 to present

Member, Standing Committee on Alternative Fuels and Technologies, Transportation Research Board, National Academies of Sciences, Engineering, and Medicine 2020 to 2028

Member, Behavior, Energy, and Climate Change (BECC) Conference Organizing Committee Recurring

Editorial and Peer Review

Reviewer for: Transportation Research Part A, Transportation Research Part C, Transportation Research Part D, Transportation Research Records, Transport Policy, Transportation, Energy Policy, Nature Energy, Journal of Transport Geography, Environment and Planning B, International Journal of Geographical Information Science, Journal of the American Planning Association, Sustainable Cities and Society, and others.

SELECTED MEDIA ENGAGEMENT

Recurring expert source for major national and international media outlets, including The New York Times, The Washington Post, The Wall Street Journal, NPR Marketplace, NPR AirTalk, CNN, Fox News, BBC, POLITICO, Wired,

MIT Technology Review, CNBC, MotorTrend, PBS NewsHour, Christian Science Monitor, Los Angeles Times, San Francisco Chronicle, Sacramento Bee, and CBS News. Made more than 60 media contributions across radio, video, and print outlets in the 2023 to 2025 review period.

Selected Recent Coverage

- Abridged / PBS KVIE, on Tesla and EV sales in Sacramento (March 2026)
- NPR Marketplace, on extended-range EVs and the U.S. market (January 2026)
- Christian Science Monitor, federal EV tax credit deadline (September 2025)
- Washington Post, "The electric-vehicle revolution may be on shakier ground than you think" (March 2025)
- CNBC, "All the Ways BYD Is Leaving Tesla in the Dust" (April 2025)
- Los Angeles Times, on Trump tariffs and California's EV ambitions (April 2025)
- New York Times, "How Toyota, a laggard on electric cars, got its fight back" (March 2024)
- Washington Post, "Plug-in hybrids vs. electric cars: We did the math" (January 2024)
- MotorTrend Channel, "Why America Is Struggling With EVs" (documentary, November 2024)
- Wired, "Automakers Are Hot for Extended-Range EVs" (November 2024)
- MIT Technology Review, "Why EVs are (mostly) set for solid growth in 2025" (January 2025)

LANGUAGES

Hebrew (native), English (fluent).